

Sintered NdFeB Magnet Grades & Specifications

High-performance magnetic parameters optimized for electric mobility, renewable generators, aerospace, and automation.

Series Classification	Grade	Remanence (B _r)		Coercivity (H _{cb})		Intrinsic Coercivity (H _{ci})		Max Energy Product (BH) _{max}	
		T	kGs	kA/m	kOe	kA/m	kOe	kJ/m ³	MGOe
Series N (Max 80°C)	N35	1.17-1.22	11.7-12.2	876	11.0	955	12.0	263-287	33-36
	N38	1.22-1.27	12.2-12.7	876	11.0	955	12.0	287-310	36-39
	N40	1.25-1.30	12.5-13.0	876	11.0	955	12.0	302-318	38-40
	N42	1.28-1.33	12.8-13.3	876	11.0	955	12.0	318-334	40-42
	N45	1.32-1.38	13.2-13.8	876	11.0	955	12.0	342-358	43-45
	N48	1.37-1.43	13.7-14.3	836	10.5	955	12.0	358-382	45-48
	N50	1.40-1.45	14.0-14.5	836	10.5	955	12.0	374-398	47-50
	N52	1.42-1.48	14.2-14.8	836	10.5	955	12.0	390-414	49-52
Series M (Max 100°C)	N35M	1.17-1.22	11.7-12.2	876	11.0	1114	14.0	263-287	33-36
	N38M	1.22-1.27	12.2-12.7	876	11.0	1114	14.0	287-310	36-39
	N40M	1.25-1.30	12.5-13.0	900	11.3	1114	14.0	302-318	38-40
	N42M	1.28-1.33	12.8-13.3	923	11.6	1114	14.0	318-334	40-42
	N45M	1.32-1.38	13.2-13.8	955	12.0	1114	14.0	342-358	43-45
	N48M	1.37-1.43	13.7-14.3	995	12.5	1114	14.0	358-382	45-48
	N50M	1.40-1.45	14.0-14.5	1027	12.9	1114	14.0	374-398	47-50
Series H (Max 120°C)	N33H	1.13-1.18	11.3-11.8	836	10.5	1353	17.0	247-263	31-33
	N35H	1.17-1.22	11.7-12.2	868	10.9	1353	17.0	263-287	33-36
	N38H	1.22-1.27	12.2-12.7	899	11.3	1353	17.0	287-310	36-39
	N40H	1.25-1.30	12.5-13.0	915	11.5	1353	17.0	302-318	38-40
	N42H	1.28-1.33	12.8-13.3	939	11.8	1353	17.0	318-334	40-42
	N45H	1.32-1.38	13.2-13.8	963	12.1	1353	17.0	342-358	43-45
	N48H	1.37-1.43	13.7-14.3	1003	12.6	1274	16.0	358-382	45-48

Series Classification	Grade	Remanence (B_r)		Coercivity (H_{cb})		Intrinsic Coercivity (H_{ci})		Max Energy Product $(BH)_{max}$	
		T	kGs	kA/m	kOe	kA/m	kOe	kJ/m^3	MGOe
Series SH (Max 150°C)	N30SH	1.08-1.13	10.8-11.3	796	10.0	1592	20.0	223-247	28-31
	N33SH	1.13-1.18	11.3-11.8	836	10.5	1592	20.0	247-263	31-33
	N35SH	1.17-1.22	11.7-12.2	868	10.9	1592	20.0	263-287	33-36
	N38SH	1.22-1.27	12.2-12.7	899	11.3	1592	20.0	287-310	36-39
	N40SH	1.25-1.30	12.5-13.0	915	11.5	1592	20.0	302-318	38-40
	N42SH	1.28-1.33	12.8-13.3	939	11.8	1592	20.0	318-334	40-42
	N45SH	1.32-1.38	13.2-13.8	963	12.1	1592	20.0	342-358	43-45
Series UH (Max 180°C)	N28UH	1.02-1.08	10.2-10.8	756	9.5	1990	25.0	207-223	26-28
	N30UH	1.08-1.13	10.8-11.3	796	10.0	1990	25.0	223-247	28-31
	N33UH	1.13-1.18	11.3-11.8	836	10.5	1990	25.0	247-263	31-33
	N35UH	1.17-1.22	11.7-12.2	868	10.9	1990	25.0	263-287	33-36
	N38UH	1.22-1.27	12.2-12.7	899	11.3	1990	25.0	287-310	36-39
	N40UH	1.25-1.30	12.5-13.0	915	11.5	1990	25.0	302-318	38-40
Series EH (Max 200°C)	N28EH	1.02-1.08	10.2-10.8	756	9.5	2388	30.0	207-223	26-28
	N30EH	1.08-1.13	10.8-11.3	796	10.0	2388	30.0	223-247	28-31
	N33EH	1.13-1.18	11.3-11.8	836	10.5	2388	30.0	247-263	31-33
	N35EH	1.17-1.22	11.7-12.2	868	10.9	2388	30.0	263-287	33-36
	N38EH	1.22-1.27	12.2-12.7	899	11.3	2388	30.0	287-310	36-39
Series AH (Max 230°C)	N28AH	1.02-1.08	10.2-10.8	756	9.5	2627	33.0	207-223	26-28
	N30AH	1.08-1.13	10.8-11.3	796	10.0	2627	33.0	223-247	28-31
	N33AH	1.13-1.18	11.3-11.8	836	10.5	2627	33.0	247-263	31-33
	N35AH	1.17-1.22	11.7-12.2	868	10.9	2627	33.0	263-287	33-36

Physical, Mechanical & Chemical Properties

Essential engineering parameters and environmental surface treatments for NdFeB system design.

Physical & Mechanical Characteristics

Density	7.4 – 7.6	g/cm³	Vickers Hardness	500 – 650	HV
Flexural Strength	200 – 300	MPa	Compressive Strength	700 – 1000	MPa
Young's Modulus	150 – 170	GPa	Electrical Resistivity	110 – 170	μΩ·cm
Thermal Conductivity	7 – 9	W/(m·K)	Curie Temperature	310 – 340	°C
Recoil Permeability (μrec)	1.03 – 1.10	ratio	Br Temp. Coefficient (α)	-0.09 to -0.13	%/°C
Hcj Temp. Coefficient (β)	-0.40 to -0.70	%/°C	Max Operating Temp (AH)	230	°C

Surface Coating & Corrosion Protection Options

Coating	Material/Type	Thickness (μm)	Salt Spray (hrs)	Max Temp	Appearance & Recommended Applications
Ni-Cu-Ni	Triple Metal Layer	10 – 25	48 – 96	200°C	Silver metallic, standard protection
Zinc (Zn)	Metallic Plating	5 – 20	24 – 72	200°C	Silver/blue, cost-effective industrial
Epoxy	Organic Polymer	10 – 40	200 – 500	150°C	Black matte, excellent chemical & moisture resistance
Ni + Epoxy	Dual Hybrid Barrier	15 – 35	500 – 1000	150°C	Black, extreme environment and saline protection
Gold (Au)	Precious Metal over Ni-Cu	1 – 3	> 1000	150°C	Gold, medical, sensor, high clean-room spec
Parylene	Polymer Vapor Deposition	5 – 30	> 500	125°C	Transparent, pinhole-free dielectric barrier
Passivation	Chemical Conversion Layer	1 – 5	< 24	250°C	Silver, temporary storage, needs dry environment
Phosphate	Inorganic Conversion	1 – 5	< 24	250°C	Grey matte, base coat for secondary gluing/assembly